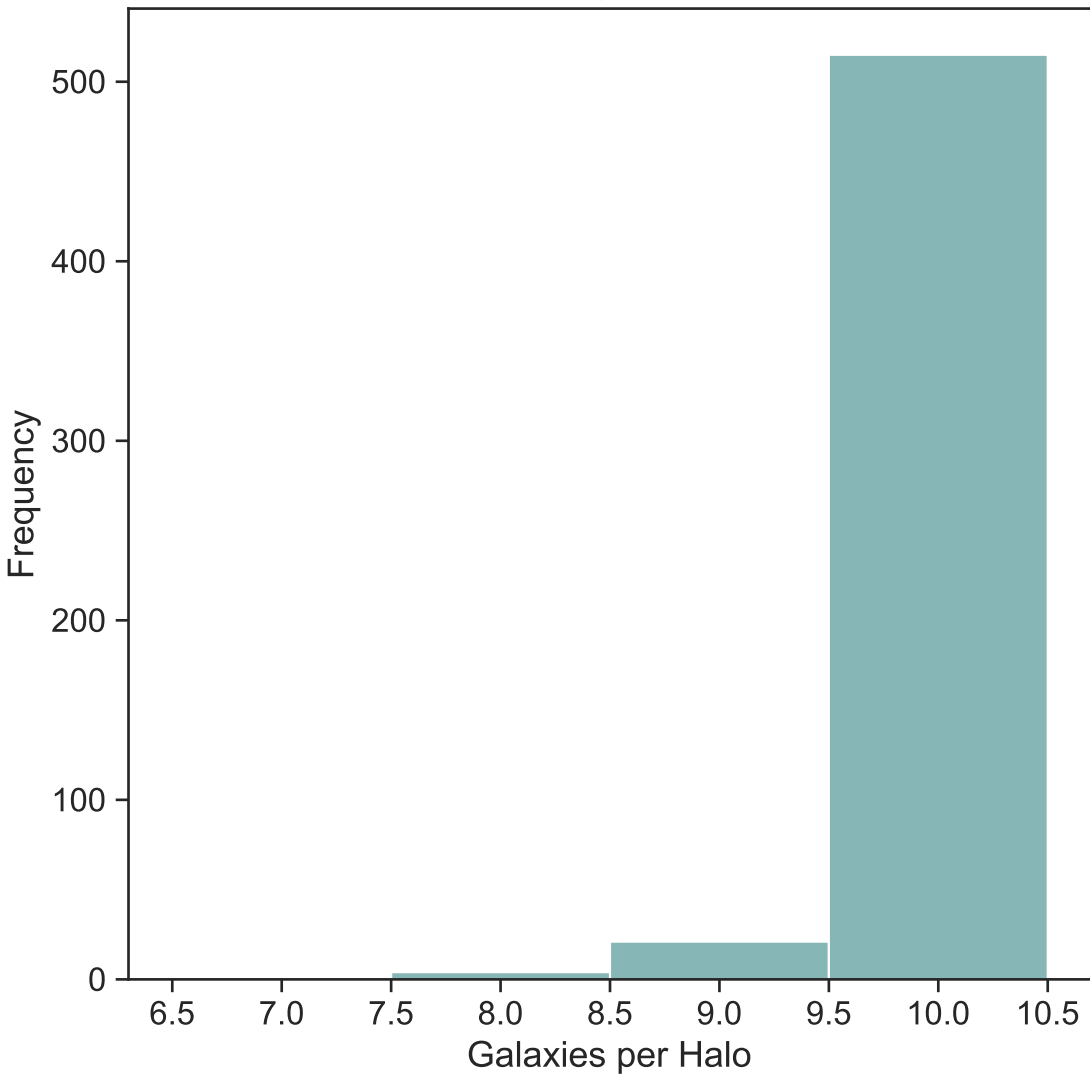
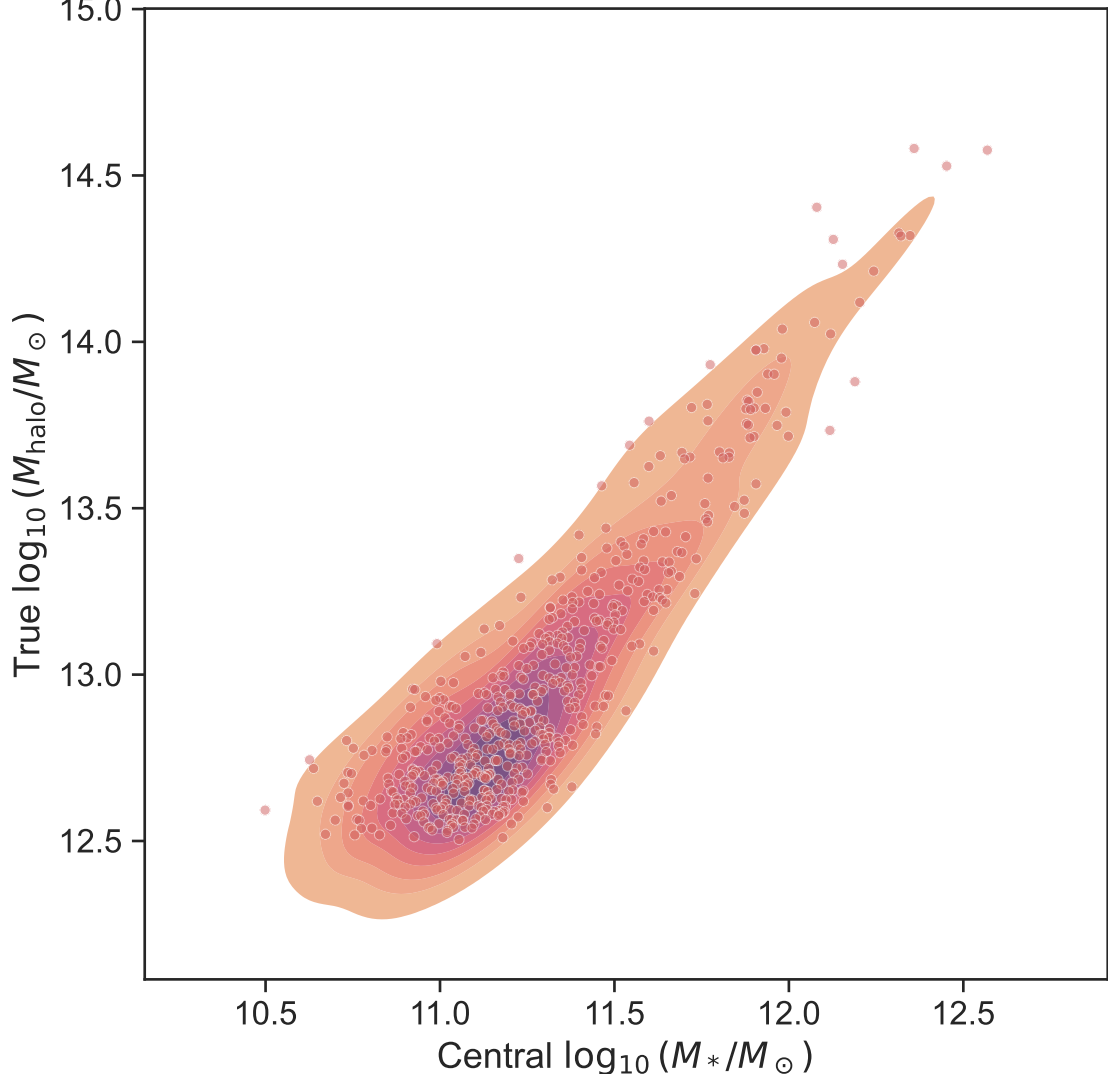


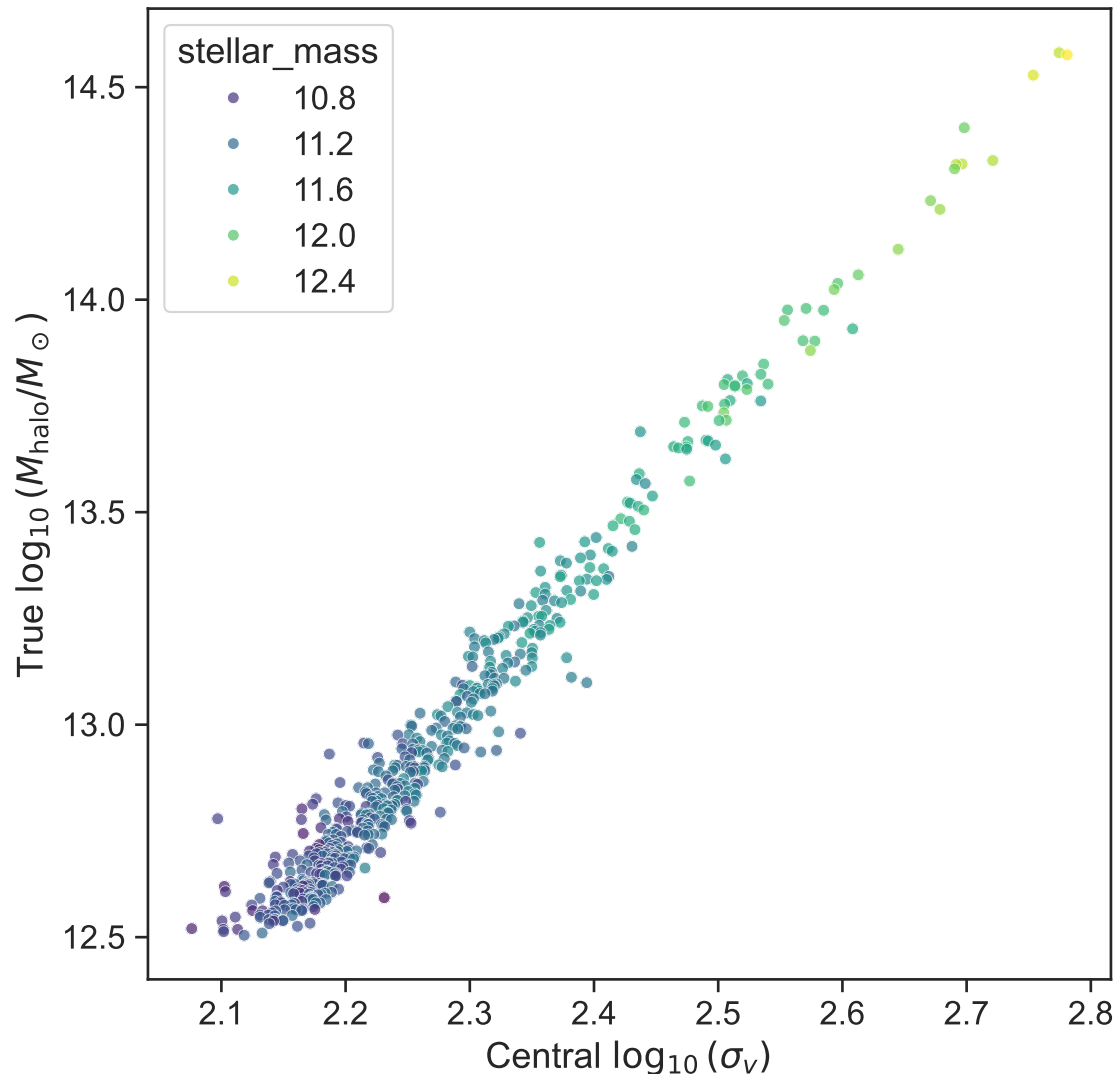
Halo Multiplicity vs Graph Size



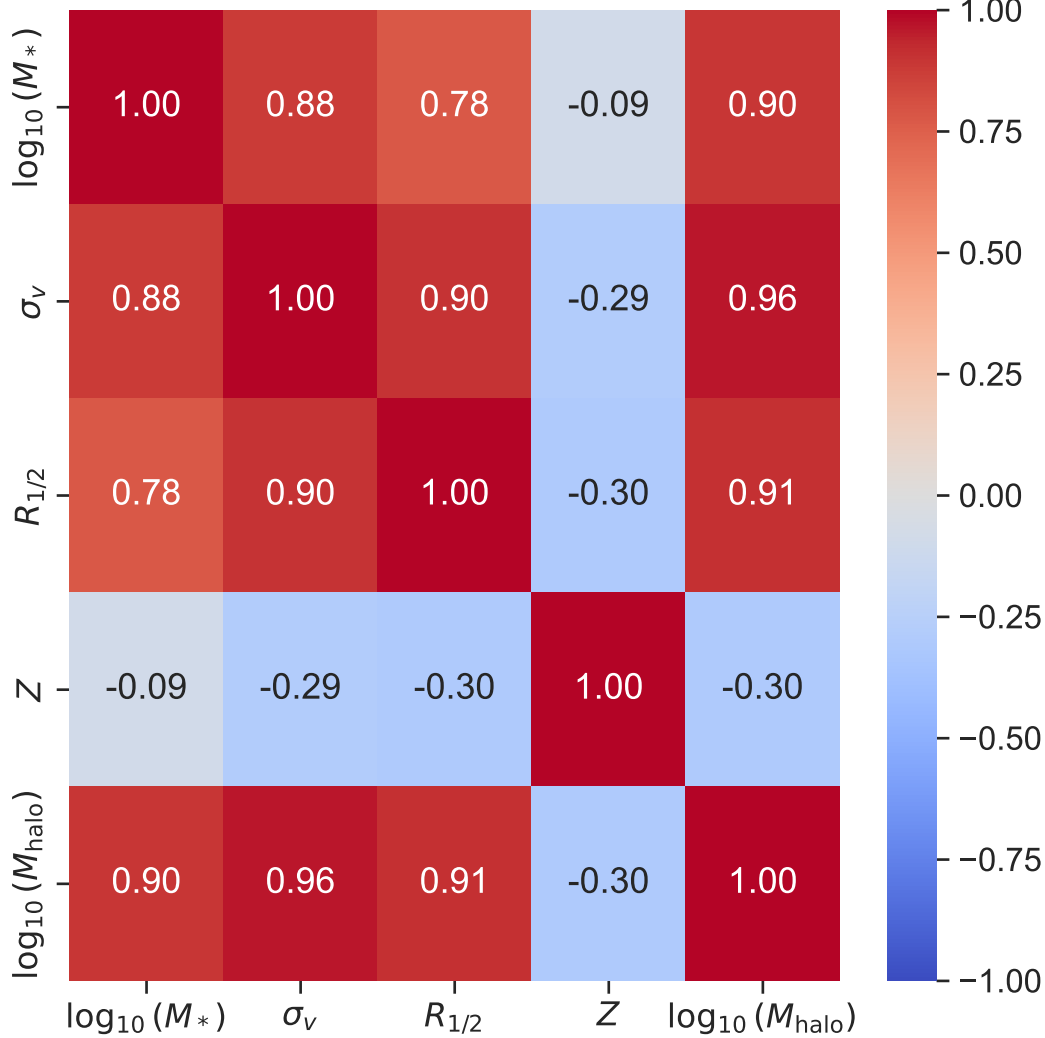
Stellar-to-Halo Mass Relation



Kinematics vs Halo Mass



Feature Linear Correlation



GNN Architecture
 $N_{\text{layers}} = 3, H = 64$
Inputs: $M_*, \sigma_v, R_{1/2}, Z$
Physics Loss: Virial Theorem ($\lambda_{\text{max}} = 0.25$)